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June 2025

A simple on-field fast knee-flexion test to assess acute knee flexor fatigue. European Journal of Applied Physiology, Ahead of Print: 1-9, 2025.

Adherence to strength training and lower rates of sports injury in contact sports: A systematic review and meta-analysis. The Orthopaedic Journal of Sports Medicine, 13(5): 1-14, 2025.

An 18-year study of changes in neurocognitive function and associations with repetitive head trauma among former collegiate American football players: A case series. Archives of Clinical Neuropsychology, Ahead of Print: 1-10, 2025.

Associations of tackling characteristics, player position, and head contact risk during game play in college football. Journal of Athletic Training, Ahead of Print: 1-26, 2025.

Combining evidence and practice to optimize neck training aimed at reducing head acceleration events in sport: A systematic review and Delphi-consensus study. British Journal of Sports Medicine, Ahead of Print: 1-16, 2025.

Consensus on the use of therapeutic interventions in pain management, recovery, and athletic performance: A Delphi study. International Journal of Sports Physical Therapy, 20(3): 420-428, 2025.

Current practice in the measurement and management of Vitamin D status in elite sport and parasport. European Journal of Sport Science, 25: e12305, 2025.

Development of a preliminary multivariable model predicting hamstring strain injuries during preseason screening in soccer players: A multidisciplinary approach. Annals of Medicine, 57(1): 2494683, 2025.

Drugs and supplements in athletic injury recovery and performance: WADA perspective. Sports Injuries: Prevention, Diagnosis, Treatment and Rehabilitation, 244-258, 2025.

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Exercise collapse associated with sickle cell trait: From consensus to confusion. British Journal of Sports Medicine, Ahead of Print: 1-2, 2025.

Growing trend in novel or experimental substances not approved for human use sold as consumer products poses threat to athletes, service members, and public health. Drug Testing and Analysis, Ahead of Print: 1-3, 2025.

Head, face, and neck cooling for performance: A systematic review and meta-analysis. International Journal of Sports Physiology and Performance, Ahead of Print: 1-21, 2025.

Identifying key factors influencing hamstring muscle strains in football (soccer) players: A narrative review with descriptive analysis. International Journal of Sport Studies for Health, 8(1): 64-78, 2025.

Impact of Nordic hamstring breaking point angle on football (soccer) player performance. PeerJ, 13: e19275, 2025.

Implicit doping attitude and athletes' accuracy in avoiding unintentional doping when being offered beverages with banned performance-enhancing substances. Journal of Science and Medicine in Sport, Ahead of Print: 1-28, 2025.

Incidence, management, and outcomes following elbow ulnar collateral ligament injuries in non-quarterback professional football players. JSES International, Ahead of Print: 1-18, 2025.

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A simple on-field fast knee-flexion test to assess acute knee flexor fatigue. European Journal of Applied Physiology, Ahead of Print: 1-9, 2025.

Purpose: In a practical setting, outside the laboratory, acute muscle fatigue may be underestimated because substantial recovery occurs during the elapsed time between the end of exercise and fatigue assessment. We introduce a simple field test to assess knee flexor contractile function quickly after exercise cessation.

Methods: Fourteen young amateur football players performed maximally fast knee flexions (FKFs) in the prone position with their dominant leg, before (pre) and 20 s after finishing a series of fourteen fatiguing 40 m sprints (post) and again following 6 min recovery (rec). Peak angular acceleration (PAA) about the knee joint was measured with a small inertial measurement unit (IMU) firmly attached to the shin.

Results: Although participants only practiced the FKFs for 1 min in the warm-up, the reliability of PPA was good with coefficients of variation of 3.0% (pre), 2.7% (post), and 3.6% (rec). Sprint time increased from 5.96 ± 0.40 s to 6.55 ± 0.37 s ($p < 0.001$, $f = 0.89$), PAA decreased from 107.1 ± 11.5 rad.s^{-2} to 94.1 ± 11.7 rad.s^{-2} ($p < 0.001$, $f = 0.50$) and following recovery ($p < 0.05$) values were 6.15 ± 0.39 s and 103.1 ± 10.7 rad.s^{-2}, respectively. The percentage decrease in PAA during FKFs was linearly related ($r2 = 0.48$, $p = 0.01$) to the percentage increase in 40 m sprint time. In addition, PAA (pre) was related to the time of the first sprint ($r2 = 0.33$, $p = 0.03$).

Conclusion: The proposed FKF test is reliable and can easily be executed to evaluate acute knee flexor muscle fatigue on the field. The presented relations between (changes in) sprint performance and peak knee angular accelerations during isolated fast knee flexions are promising but need confirmation in larger-scaled studies.

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Adherence to strength training and lower rates of sports injury in contact sports: A systematic review and meta-analysis. The Orthopaedic Journal of Sports Medicine, 13(5): 1-14, 2025.

Background: Injury prevention is very important for athletes, and studies have been carried out to better inform effective means of injury prevention. Strength training has long been recognized as important for preventing sports injuries. But until now, there has been debate about the proper application of strength training to prevent injuries, especially for different types of injury.

Purpose: To compare the effects of single-component and multicomponent strength training on the prevention of different types of injuries and to provide reference for the future design of injury prevention programs (IPPs).

Study Design: Systematic review; Level of evidence, 1.

Methods: Five databases were searched in August 2024. The main inclusion criteria were randomized controlled trials on use of an IPP that included strength training with a primary outcome of sports injury rate. The initial search resulted in 3583 articles, which were filtered to 16 articles that met the inclusion criteria. Extracted data were subjected to meta-analysis using a random-effects model, and subanalysis was carried out to explore the efficacy of different component strength IPPs in mitigating injuries at different locations in the body.

Results: The pooled results based on total injuries showed that strength IPPs had a statistically significant reduction in injury relative risk (RR) of 0.70 (95% CI, 0.60-0.82). Subgroup analysis showed that single-component strength training significantly reduced groin injuries (RR, 0.69; 95% CI, 0.51-0.93) and hamstring injuries (RR, 0.37; 95% CI, 0.25-0.55). In addition, multicomponent strength training significantly reduced knee injuries (RR, 0.71; 95% CI, 0.51-0.98) and ankle injuries (RR, 0.68; 95% CI, 0.52-0.89).

Conclusion: This study demonstrates that different-component strength IPPs have differential preventive effects on sports injury because of different target sites. The reason may be that the anatomic structure and motor function of the target sites are different. We can determine the

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vulnerable locations according to the different characteristics of the sports and select different component strength IPPs according to the characteristics of the target sites.

An 18-year study of changes in neurocognitive function and associations with repetitive head trauma among former collegiate American football players: A case series. Archives of Clinical Neuropsychology, Ahead of Print: 1-10, 2025.

Objectives: To assess changes in neurocognitive test scores among former collegiate American football players over 18 years and examine associations with head trauma history.

Methods: Former collegiate American football players (n = 31; aged = 38.4 ± 1.3 years) self-reported their concussion history and repetitive head impact exposure (Head Impact Exposure Estimate). Neurocognitive testing was conducted at two time-points (during college [T1] and 18-year follow-up [T2]) via Hopkins Verbal Learning Test-Revised Immediate and Delayed Recall; Verbal fluency; Symbol Digit Modalities Test; and Trail-Making Test-B. Raw score changes were calculated, with accompanying Cohen's d effect sizes and coefficients of variation. Repeated-measures analyses of covariance models were fit to examine changes in test performance. Multivariable linear regression models tested associations of sport-related concussion history and repetitive head impact exposure with change scores.

Results: No significant changes in cognitive test performance were observed (ps ≥ .06). Individual-level changes exhibited high variability (coefficients of variation ≥ 244%), and group-level effects were small to medium (d ≤ 0.41). Neither sport-related concussion history nor repetitive head impacts were related to change scores (ps > .05).

Conclusions: Group-level test scores did not change over 18 years among former collegiate football players now in midlife, though individual-level variability was high. Sport-related concussion and head impact exposure estimates were not related to change. Longitudinal studies are essential to understand cognitive trajectories of former football players and factors influencing those trajectories.

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Associations of tackling characteristics, player position, and head contact risk during game play in college football. Journal of Athletic Training, Ahead of Print: 1-26, 2025.

Context: Sport-related concussion is a common injury among National Collegiate Athletic Association football athletes. Beginning with the 2016 season, Ivy League Conference coaches voted to ban player-on-player tackling from all in-season practices. BLINDED have enforced a no-tackle approach in practices since 2010.

Objective: To examine the association between tackling technique and head contact risk, and compare base rates of techniques used in the 2016 season between an Ivy League team with a longstanding no-tackle practice policy vs. the rest of the league.

Design: Cross-sectional study.

Setting: Ivy League College Football Conference.

Patients or other participants: Two-hundred-thirty-seven Ivy League defensive football players that participated in the 2016 season.

Main outcome measure(s): Tackles (N=3,701) across 237 Ivy League defensive football players in the 2016 season were coded based on predetermined classifications, which were combined to create unique tackle combinations/techniques. Associations among tackling techniques, head impact risk, and team (BLINIDED vs. other Ivy League teams) were evaluated using logistic regression, yielding odds ratios (OR) for head contact.

Results: Low-risk tackle characteristics for head contact during a tackle were neutral neck position (OR=0.1), back contact area (OR=0.3), pursuing momentum (OR=0.5), and quarterback sack momentum (OR=0.3). Low-risk tackle techniques were high-back-neutral (OR=0.1), low-back-neutral (OR=0.2), and medium-back-neutral (OR=0.1). High-risk tackle characteristics were flexion (OR=14.2) and extension (OR=3.8) neck positioning, front contact (OR=2.2), blowup (OR=2.5), and cut (OR=3.0). High-risk tackle techniques included low-side-flexion (OR=4.9), low-front-flexion (OR=9.9), medium-side-flexion (OR=15.5), and medium-front-flexion (OR=11.4). Relative to BLINDED, other teams demonstrated higher odds of using high-risk techniques (low-side-flexion OR=3.5; low-front-flexion OR=3.9; medium- side-

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flexion OR=6.3; medium-front-flexion OR=2.3) and reduced odds of using low-risk tackle combinations (high-side-neutral OR=0.4; high-back-neutral OR=0.6; medium-side-neutral OR=0.8).

Conclusions: Tackling techniques are associated with head contact risk, and by extension, player safety. BLINDED, who have a longstanding policy of practicing without player-on-player tackling, showed reduced use of high-risk tackling techniques.

Combining evidence and practice to optimize neck training aimed at reducing head acceleration events in sport: A systematic review and Delphi-consensus study. British Journal of Sports Medicine, Ahead of Print: 1-16, 2025.

Abstract - Head acceleration events (HAEs) can potentially have adverse consequences for athlete brain health. In sports, in which head injuries have the highest incidence, identifying strategies to reduce HAE frequency and magnitude is a priority. Neck training is a potential strategy to mitigate against the magnitude of HAEs. This two-part study aimed to (1) systematically review the literature of neck training interventions in sport and (2) undertake an expert Delphi consensus on the best practices for neck training implementation to reduce HAEs in sport. Part I: a systematic search of four databases was undertaken from the earliest records to September 2024. The PRISMA (Preferred Reporting Items for Systematic Review and Meta-Analysis) guidelines were followed, and a quality assessment was completed using a modified Downs and Black assessment tool and the GRADE (Grading of Recommendations Assessment, Development and Evaluation). Papers were eligible if they both (1) implemented a reproducible exercise intervention targeting the neck within collision, combat or motor sport, and (2) assessed outcomes relating to either: the physical profile of the neck; head/neck injury incidence; and/or HAEs. Part II: 18 international experts, with experience in research and/or applied practice of neck exercise training, concussion and/or HAEs, reviewed the part I findings before completing a three-round Delphi consensus process. Part I included 21 papers, highlighting the heterogeneity of existing interventions. Part II resulted in 57 statements coded into five categories: contextual factors (n=17), neck training periodization (n=12), training adaptations (n=10), neck training content (n=15) and athlete adherence (n=3). This study presents recommendations for neck exercise training aiming to reduce HAEs in sport, supporting both practice and future research.

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Consensus on the use of therapeutic interventions in pain management, recovery, and athletic performance: A Delphi study. International Journal of Sports Physical Therapy, 20(3): 420-428, 2025.

Background: Therapeutic interventions are widely used in clinical practice to prepare individuals for movement, recover from exercise, and mediate pain and inflammation associated with injury. However, there has yet to be cited consensus regarding when to use such interventions.

Purpose: Using the Delphi method, this study aimed to develop consensus among physical therapists (PTs) and athletic trainers (ATs) regarding the use of percussive, vibration, infrared, heat, and cryotherapy in a variety of treatment circumstances. These included preparation and recovery from physical activity, treatment of acute and chronic joint pain, and treatment of acute and chronic soft tissue/muscle pain.

Methods: Four rounds of surveys were delivered to individuals who were licensed or certified PTs/ATs, with more than 1,000 hours of experience working with athletic populations and experience using at least one of the interventions of interest in the prior five years. Consensus was defined as a greater than 70% agreement, and consideration for use was defined as selecting "In Some Instances'' and"In Most Instances."

Results: Ninety-four individuals responded to the screening survey (n1 = 74, n2 = 68, n3 = 44, n4 = 32). The individuals who participated agreed that they would consider using the following therapeutic interventions in treatment: 1) percussive therapy, local vibration, and local heat therapy in the preparation for physical activity; 2) local cryotherapy in recovery from physical activity; 3) local cryotherapy for the treatment of acute joint pain and acute soft tissue/muscle pain; 4) local heat and local cryotherapy for the treatment of chronic joint pain; and 5) percussive, local vibration, local heat, and local cryotherapy in the treatment of chronic soft tissue/muscle pain.

Conclusion: Clinical intervention often relies on patient presentation and preference. This study provides a consensus on the use of common therapeutic interventions for the management of athletic recovery, pain, and inflammation among ATs and PTs.

Level of evidence: 3.

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Current practice in the measurement and management of Vitamin D status in elite sport and parasport. European Journal of Sport Science, 25: e12305, 2025.

Abstract - The field of vitamin D research in sport has stagnated, with a notable lack of new evidence challenging existing paradigms. Despite well-documented consequences of vitamin D deficiency in athletes, gaps remain in understanding the feasibility of best practices for measuring and managing vitamin D status in elite sports. This survey aimed to define whether practitioners test athletes' vitamin D status, the methods used, decision-making regarding supplementation and sources of information on vitamin D. Seventy-four stakeholders from 26 sports and parasports participated, representing athletes of eight ethnicities across the United States, United Kingdom, Europe, Australia and Asia. Financial and logistical constraints were significant barriers to effective vitamin D testing and management. Testing practices varied widely, with venous blood sampling being the most common method. Many practitioners were unaware of the specific analytical methods used. Supplementation decisions were typically collaborative, involving the sport science support team, but approaches and criteria varied considerably among practitioners. Most sourced information were from academic literature and consultations. These results are the first to characterize the perceptions and practices of practitioners in elite sport and parasport regarding vitamin D testing and supplementation. Despite extensive literature on the vitamin D status of athletes and its impact on performance, our findings indicate stagnation in innovative practices for optimizing vitamin D status. Technological improvements to reduce testing costs and collaborative approaches between practitioners and researchers could facilitate knowledge delivery and improve practices.

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Development of a preliminary multivariable model predicting hamstring strain injuries during preseason screening in soccer players: A multidisciplinary approach. Annals of Medicine, 57(1): 2494683, 2025.

Objective: Reducing the incidence of hamstring strain injuries (HSIs) is a priority for soccer clubs. However, robust multifactorial predictive models are lacking and potential predictors such as sprint kinematics, performance fatigability, and psychological variables have been overlooked. Thus, the aim of this study was to develop a preliminary parsimonious multifactorial model to predict players at risk of HSI through preseason screening.

Materials and method: Psychological, physiological, kinematic, performance fatigability and health-related variables were collected for 120 regional and national soccer players during the 2022 preseason. HSIs were prospectively recorded over the entire soccer season. After variable selection, logistic regressions with the Wald backward stepwise method were used to refine the model. The predictive abilities of the model and of the individual variables were determined using the area under the receiver operating characteristic curve (AUC).

Results: Twenty-nine players sustained an HSI during the follow-up period. The final model included eight variables: age, sex, HSI history, knee flexor performance fatigability, sprint performance (best sprint time and maximal theoretical velocity V0), perceived vulnerability to injury, and subjective norms in soccer. While its model was preliminary, it showed good fit indices and strong predictive performance (true positive rate: 79%, AUC = .82). None of the variables evaluated independently demonstrated satisfactory performance in predicting HSI (AUC≤.65).

Conclusion: Using a multidisciplinary approach and measurements of only a few variables during preseason screening, the current model tends to demonstrate high accuracy in identifying soccer players at risk of HSI.

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Drugs and supplements in athletic injury recovery and performance: WADA perspective. Sports Injuries: Prevention, Diagnosis, Treatment and Rehabilitation, 244-258, 2025.

Abstract - The use of drugs and nutritional supplements to enhance sports and exercise performance has been widespread at all levels of participation and competition, albeit to a varying extent. The World Anti-doping Agency (WADA) was established to address the problem of drug use by athletes for their potential ergogenic effects. The primary purpose of WADA is to protect athletes worldwide, promote the values of clean sport, and preserve the spirit of sport. WADA's primary role is to develop, harmonize, and coordinate anti-doping rules and policies across all sports and countries. In addition, WADA engages in a wider range of other activities including scientific and social science research; education; intelligence and investigations; development of anti-doping capacity; and monitoring of compliance with the World Anti-Doping Program. The detection of a prohibited substance or its metabolite or laboratory markers in an athlete's sample (urine, blood, or other as appropriate) constitutes doping. The categories of WADA prohibited substances and methods include anabolic agents, peptide hormones and their releasing factors, growth factors and growth modulators, erythropoiesis promoting agents and methods, beta-2 agonists, hormone and metabolic modulators, and masking agents such as diuretics. Optimal baseline nutrition, treating any micronutrient deficiencies, and maintaining appropriate fluid and electrolyte balance are all essential for optimal sports performance. Although a large number of nutritional supplements are being marketed for their purported ergogenic effects, only a handful have been shown to have such effect to a variable degree; these include caffeine, creatine, nitrate, beta-alanine, and sodium bicarbonate.

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Exercise collapse associated with sickle cell trait: From consensus to confusion. British Journal of Sports Medicine, Ahead of Print: 1-2, 2025.

Overview - Sickle cell trait (SCT) and its association with exertional rhabdomyolysis (ER) and exercise-related death (ERD) in warfighters and athletes (WA) are significant and controversial issues confronting sports medicine communities and the US Department of Defense (DoD). Although SCT carrier status is largely considered a benign condition, potential risks, in particular with intense repetitive or continuous exertion, are increasingly recognized. The deaths of two soldiers in 2010 prompted the convening of a summit by the Consortium for Health and Military Performance at the Uniformed Services University in Bethesda, Maryland. In collaboration with the American College of Sports Medicine, interested DoD entities, the American Medical Society for Sports Medicine, the American Society of Hematology and selected members of the haematology and sports medicine community met to discuss SCT with particular emphasis on mitigating risk in WA. During that summit, the term 'exercise collapse associated with sickle cell trait' (ECAST) was adopted, recognizing the increased relative risk of both ER and ERD in WA, in the absence of a clearly recognized pathophysiology. A subsequent series of ECAST events in military personnel in 2019 prompted a reevaluation of current efforts and called for a second consensus conference. As a result of that conference, the US Army adopted a policy for SCT screening and education to align with all other DoD services. Consensus management strategies included identification of the struggling warfighter or athlete, with urgent medical assessment to include the provision of oxygen, the availability of an automated external defibrillator (AED) and a low threshold for transfer to a medical facility prepared to manage a potential fulminant metabolic crisis. Additionally, further basic science and clinical research was endorsed as an urgent need to provide stronger evidence-based guidance.

Growing trend in novel or experimental substances not approved for human use sold as consumer products poses threat to athletes, service members, and public health. Drug Testing and Analysis, Ahead of Print: 1-3, 2025. *No abstract available*

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Head, face, and neck cooling for performance: A systematic review and meta-analysis. International Journal of Sports Physiology and Performance, Ahead of Print: 1-21, 2025.

Purpose: Cooling the head, face, and neck can have strong perceptual effects that contribute to improved performance. This systematic review aimed to determine the effect of cooling strategies targeting the head, face, and neck on physical and cognitive performance, determine any associated physiological and perceptual responses, synthesize adverse events, and provide practical applications.

Methods: We conducted a systematic review and multilevel meta-analysis, adhering to PRISMA guidelines. Studies that investigated the effect of cooling strategies targeting the head, face, or neck on a physical or cognitive task using a controlled trial design were included.

Results: Sixty-three studies were identified, involving 618 participants (86.6% male). Cooling strategies included water-perfused devices (18.7%), phase-change neck collars (17.3%), fanning/cold air (14.7%), phase-change headwear (13.3%), ice/gel packs (13.3%), cold towels (5.3%), menthol application (4.0%), water spraying/dousing (4.0%), or a combination of strategies (9.3%). The effect of cooling on both self-paced and fixed-intensity exercise tasks was inconclusive; the 95% CI of the pooled effect was compatible with no effect and medium beneficial effects but not harmful effects. We were unable to pool cognitive data. Cooling reduced the skin temperature at the target site and improved thermal sensation and comfort. Effects on heart rate and core and mean skin temperatures were negligible. Adverse events were rare, and no intervention subgroup was superior.

Conclusion: We recommend that athletes experiment with a range of head-, face-, and neck-cooling strategies, including using different doses and timings, to determine the optimal strategy for their individual and sport context.

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Identifying key factors influencing hamstring muscle strains in football (soccer) players: A narrative review with descriptive analysis. International Journal of Sport Studies for Health, 8(1): 64-78, 2025.

Objective: This study aimed to explore the key intrinsic and extrinsic factors contributing to hamstring muscle strains in football players.

Methods and Materials: A narrative review approach was employed, synthesizing findings from peer-reviewed articles published between 2010 and 2024. A descriptive analysis was conducted to evaluate trends, gaps, and variations in the literature on hamstring strains. Data were extracted from studies focusing on intrinsic factors (e.g., muscle imbalances, flexibility, injury history) and extrinsic factors (e.g., training load, playing conditions, coaching methods), as well as prevention and rehabilitation approaches, including eccentric strengthening, neuromuscular training, and workload management.

Findings: The review revealed that intrinsic factors such as muscle strength imbalances, limited flexibility, and prior injury history are significant contributors to hamstring strains, while extrinsic factors, including high training loads, artificial playing surfaces, and inadequate coaching practices, exacerbate injury risk. Evidence-based interventions, such as Nordic hamstring exercises, dynamic warm-ups, and load monitoring, effectively reduce injury rates and recurrence. However, gaps in the literature were identified, including the role of genetic predisposition and the long-term impact of rehabilitation programs. Variations in the implementation and adherence to prevention strategies were noted across teams and competitive levels.

Conclusion: Hamstring strains in football are a multifactorial issue requiring a holistic approach to prevention and rehabilitation. Addressing both intrinsic and extrinsic factors through individualized and evidence-based strategies can significantly reduce injury incidence and recurrence.

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Impact of Nordic hamstring breaking point angle on football (soccer) player performance. PeerJ, 13: e19275, 2025.

Background: Football demands both aerobic and anaerobic capacities due to its dynamic movements, which include jumps, directional changes, ball control, and sprints. The Nordic hamstring exercise (NHE) enhances eccentric strength, which is crucial for high-intensity movements. However, the relationship between Nordic breaking point angle (NHEbpa), which is associated with eccentric hamstring peak torque during Nordic hamstring exercise, and sprint, countermovement jump (CMJ), and change of direction (COD) speed in soccer players has not been sufficiently investigated.

Objective: This analysis examines the relationship between the Nordic breaking point angle and critical performance indicators-sprint speed, COD ability, and vertical jump-in football players.

Methods: Fifty-eight male soccer players volunteered for the study. Assessments included anthropometric measurements, CMJ tests, 10-20-30 m sprint tests, and COD (zig-zag) tests. NHEbpa was measured using motion analysis software. Correlation analysis was used to determine the relationship between variables. A multiple linear regression analysis was conducted to evaluate the individual effects of the sprint, CMJ, and COD performances on NHEbpa, with data analyzed using JASP 0.18.

Results: Correlation analysis showed strong positive correlations between NHEbpa and sprint performances ($r = 0.633$ to 0.666), moderate negative correlation between NHEbpa and CMJ ($r = -0.406$), and moderate positive correlation between NHEbpa and COD ($r = 0.580$). Regression analysis results were used to analyze the independent coefficients of multiple variables more comprehensively, revealing significant predictors for performance: 20-m sprint ($\beta = 24.166$, $p = 0.030$), 10-m sprint ($\beta = 22.564$, $p = 0.047$), 30-m sprint ($\beta = 10.677$, $p = 0.027$), and CMJ ($\beta = 4.974$, $p = 0.034$). Conversely, COD performance ($\beta = -0.154$, $p = 0.470$) did not demonstrate a significant effect.

Conclusions: The study identified significant relationships between NHEbpa and sprint/CMJ performances ($p < 0.05$), while no meaningful effect was observed for COD speed ($p > 0.05$). These findings highlight the importance of eccentric strength in sprint performance, suggesting that other factors may play a more prominent role in COD. Incorporating eccentric-focused training, particularly through Nordic exercises, is recommended to enhance sprint performance and hamstring strength, which are essential for football players.

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Implicit doping attitude and athletes' accuracy in avoiding unintentional doping when being offered beverages with banned performance-enhancing substances. Journal of Science and Medicine in Sport, Ahead of Print: 1-28, 2025.

Objectives: Implicit doping attitude reflects the strength of one's automatic or unconscious evaluation of doping. This cross-societal study examined how implicit doping attitude predicted athletes' accuracy in avoiding unintentional doping.

Design: A real-time experimental design with cross-sectional data from three geographical regions.

Methods: A total of 681 athletes (Mage = 28.21, SD = 8.43; female = 47.1 %) from Hong Kong (N = 177), Australia (N = 164) and the United Kingdom (N = 340) completed two real-time experimental tasks for measuring their implicit doping attitude (a brief single-category implicit association test) and the accuracy of avoiding unintentional doping (a novel canned beverage sorting task).

Results: Implicit doping attitude was significantly and negatively ($\beta = -0.09$, $p < .05$, $R2 = 0.03$, 95 % CI of $\beta = -1.64$ to -0.15) associated with athletes' accuracy in avoiding unintentional doping by screening out beverages with banned performance-enhancing substances. This association was maintained when we statistically controlled for the effects of society.

Conclusions: Athletes with a positive implicit doping attitude were less accurate in determining whether unknown beverages with the possible presence of banned performance-enhancing substances should be consumed. The negative association between implicit doping and athletes' accuracy in avoiding unintentional doping appeared to be consistent across societies.

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Incidence, management, and outcomes following elbow ulnar collateral ligament injuries in non-quarterback professional football players. JSES International, Ahead of Print: 1-18, 2025.

Background: Ulnar collateral ligament (UCL) injuries can cause significant disability, especially in overhead throwing athletes. A majority of studies in the literature focus on the effects of UCL injuries in baseball players, but little is known about the severity and outcomes of these injuries in elite non-throwing contact athletes. The purpose of this study was to characterize the severity of UCL injuries and describe management outcomes in non-throwing professional football players.

Methods: A retrospective review was conducted using the National Football League (NFL) Injury Surveillance System (ISS) on non-quarterback players that had a UCL injury from 2015-2022. Player demographics, player position, severity and mechanism of injury, type of management, and time to return to play (RTP) were collected. The severity of injury was classified based on laxity on clinical examination and graded on a scale of I-III. Data was analyzed to determine the number of injuries that occurred per year, per position, and per setting to determine differences in time from play that was missed based on injury severity. Statistical significance was set a $p < 0.05$.

Results: A total of 268 non-quarterback NFL players sustained isolated UCL injuries between 2015-2022, including 33.5±4.2 injuries per year. Offensive and defensive linemen were the most common positions to sustain isolated UCL injuries, accounting for 56.0% (150/268 total injuries total) of all non-quarterback UCL injuries in the NFL. Ninety-eight percent of player with isolated UCL injuries were treated non-operatively with only 4 players requiring surgical treatment of their injury. Seventy-three percent of injuries were Grade 1 sprains, 16.8% were Grade II sprains, and 10.1% were Grade III sprains. Seventy-two percent of athletes did not miss any time due to their isolated UCL injury. In the 28% of players that did miss time, the mean-time missed was 5.4±24.2 days. There was a significant difference in the time missed between Grade I and Grade II injuries (1.1 days vs. 12.4 days; $p < 0.05$) and Grade I and Grade III injuries (1.1 days vs. 26.3 days; $p < 0.05$).

Conclusions: Isolated UCL injuries are relatively uncommon injuries in non-quarterback NFL players. The vast majority of injuries are Grade I sprains and are treated non-operatively. Grade II and Grade III injuries result in significantly increased time missed compared to Grade I injuries. This information can be useful when counseling players regarding prognosis and timing of RTP based on severity of injury.

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Influence of travel demands and match load on recovery following post-match travel in National-team footballers. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2025.

Purpose: This study investigated the relationship between travel demands and match loads on perceptual recovery, fatigue, and sleep following postmatch travel in national football teams. In addition, the influence of travel demands and the time between match kickoff and travel departure on postmatch recovery was examined.

Methods: Match-running load (via GPS) and travel data were obtained from 79 male national-team footballers. Postmatch travel duration, direction, context, time-zone difference, and time between kickoff and travel departure were collated. Athletes provided perceptual ratings of fatigue, soreness, sleep, stress, and recovery from 1 day prematch through to 3 days after postmatch travel. Linear mixed models assessed the influence of match load and travel on perceptual ratings for 3 days posttravel. Additional models assessed a standardized postmatch time point of match day + 3 to determine whether timing and extent of travel influenced recovery.

Results: Higher match loads were associated with poorer recovery, fatigue, soreness, and sleep (P = .001–.032). Athletes reported poorer fatigue, soreness, and recovery when traveling from national teams back to clubs compared with between national-team matches (P < .001). Traveling eastward was associated with poorer sleep (P = .004). Longer periods between kickoff and travel departure were associated with poorer sleep on match day + 3 (P = .003).

Conclusions: Perceptual recovery, fatigue, sleep, and soreness following postmatch travel were affected by both match load and travel demands. Greater match loads and eastward postmatch travel may impair recovery. In addition, departing later after a match was associated with poorer sleep on match day + 3; however, no relationships existed to any other recovery measures.

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Integrative field-based health and performance research: A narrative review on experimental methods and logistics to conduct competition and training camp studies in athletes. Sports Medicine, Ahead of Print: 1-27, 2025.

Abstract - Field-based sport research involves studies that collect data from athletes and/or teams during competition and/or their daily training environments. Over the last decade, sport-specific field-based research projects have significantly increased in number and complexity, partially owing to the further development of more portable measurement equipment (e.g., indirect calorimetry, desktop blood/gas analyzers, portable laboratories, etc.) and/or wearable or consumable technologies (e.g., smart watches, sensors, core temp pills, etc.). However, given these rapid advances and novelty, challenges remain in the validity and applicability of these devices. Unfortunately, there are no global ethical or best-practice standards for the use of portable devices and/or wearables in sport; however, this review will outline various opportunities and challenges. Many decision trade-offs are required when designing field-based research studies to balance gold-standard scientific rigor and strict research control with highly applied, but less-controlled, "real-world" conditions. To our knowledge, there are no narrative reviews that take a wholistic view of the logistical and methodological considerations of field-based research in athletes. Accordingly, this review takes a multi-disciplinary methodological approach (physiological, nutritional/energetic, biomechanical, musculoskeletal, cognitive, and psychosocial factors), along with the logistical considerations involved in project planning, research design, and ethics of field-based research with elite athletes and/or teams. We also provide practical guidance for characterizing the extreme demands of elite training and competition to support research that ultimately catalyzes improved understanding of the limits of human capacity. We hope this review can serve as a practical guide for researchers undertaking elite athlete field-based research.

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Mixing the sheep and the goats: Reframing the interpretation of acceleration data in sport. International Journal of Sports Physiology and Performance, Ahead of Print: 1-4, 2025.

Purpose: Acceleration is a continuous variable, and current practices of categorizing it into nonconsensual arbitrary thresholds may potentially lead to imprecise and biased data interpretation, as well as limiting interstudy comparisons. This commentary proposes a reframing of the interpretation of acceleration data and their use for athlete monitoring and exercise prescription. Although this applies to multiple sports, we will focus on the example of team sports in which these metrics are commonly used.

Conclusion: Despite the recognized benefits of acceleration monitoring, 2 major limitations are evident. First, the common practice of classifying continuous acceleration data into discrete categories leads to a loss of relevant information, thus reducing the validity, reliability, and statistical power of analyses. Second, the use of arbitrary thresholds fails to account for individualized profiles, which are essential for monitoring. The limitations of these approaches are exacerbated by the lack of any objective justification for the thresholds used. We propose that acceleration be analyzed continuously, and, therefore, interpretation of data should be individualized. Accelerations should be analyzed as continuous variables rather than categorized into arbitrary thresholds. Individualized monitoring using players' maximum acceleration as a reference can provide a more accurate representation of players' physical demands.

No association between fencing response after concussion and recovery time, concussion severity, or clinical outcomes in professional American football players. Neurosurgery, Ahead of Print: 1-8, 2025.

Background and objectives: The relationship between tonic posturing (also known as the "fencing response") after concussion and clinical outcomes is unclear. The objective of this study was to compare clinical outcomes between professional American football players with a documented fencing response after concussion to players without documented fencing response after concussion.

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Methods: Players with a fencing response after concussion were matched 1:2 to players without documented fencing response after in-game concussion by age, position, and concussion history from the 2018 to 2023 seasons. The primary outcome measure was number of days until return to full participation. Secondary outcome measures included screening of neurocognitive function with the Standardized Assessment of Concussion (SAC) and modified Balance Error Scoring System (mBESS), and subjective symptom reporting with the Sport Concussion Assessment Tool-5. An independent-samples median test compared game-day SAC and mBESS outcomes between groups. A Kaplan-Meier curve evaluated fencing response after concussion on days until return to full participation. A linear mixed-effects model for 11 Sport Concussion Assessment Tool-5 total symptom scores (ie, day of injury up to postinjury day 10) compared symptom recovery over time between fencing groups.

Results: Fifty-seven fencing responses were documented over 6 seasons. Nearly 2-in-3 players with a fencing response played defensive back or wide receiver. There was no difference between players with a fencing response (median = 10 days) and those without fencing response (median = 9 days) in days until return to full participation (P = .16). Fencing response was not associated with a change in hazard for days until return to full participation ($\chi 2 = 0.91$; P = .34). There was no difference between groups in SAC scores (P = .18-.99) or mBESS score (fencing: 4, nonfencing: 4; P = .87). There was no difference between groups in symptom scores over time (β: -0.02; P = .92).

Conclusion: Fencing response after concussion was not associated with symptom recovery, concussion severity, or short-term recovery in professional American football.

Nutrition for optimizing immune function and recovery from injury in sports. Clinical Nutrition ESPEN, 66: 101-114, 2025.

Abstract - The immune system, sports injuries, physical activity and clinical sports nutrition are closely linked. Inadequate nutrition and intense stress in athletes increase the risk of immune dysfunction, injury and disease, while injury reduces the positive effects of physical activity, creating a vicious cycle affecting health and performance. Nutritional support is key to reducing the risk of injury, speeding up recovery and shortening rehabilitation. The intake of energy, protein, carbohydrates, fats and micronutrients should be adapted to the needs following injury. Quality nutrition is preferred over the

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use of dietary supplements, which are limited to cases of identified deficiencies (e.g. vitamin D or iron deficiencies). Health professionals with knowledge of clinical nutrition have a key role to play in injury prevention, treatment and rehabilitation, and nutritional interventions are an indispensable part of a holistic approach to athletes' health.

On-field kinematics of cut maneuvers in football players: Are wearable sensors reliable for assessing anterior cruciate ligament injury risk? Journal of Sports Sciences, Ahead of Print: 1-12, 2025.

Abstract - The aim of the present study was to present the reliability and normative values of wearable inertial sensors kinematics during football-specific cut maneuver tasks collected on the football field for assessing anterior cruciate ligament (ACL) injury risk. Forty-seven academy football players (age 15.9 ± 2.4 years, female n = 20) performed a planned 90° change of direction within the Agility T-test and unplanned football-specific changes of direction (FS deceiving action). Kinematics was collected through eight wearable inertial sensors (100 hz, MTw Awinda, Movella). Intraclass correlation coefficient and Root Mean Square Error were used to inspect test-retest and side-to-side reliability of peak and waveform kinematics. Normative kinematics was compared between male and female players (t-test with Cohen's d, $p < 0.05$). Test–retest reliability was moderate-to-excellent in most of the parameters (r=0.40–0.92). Side-to-side reliability was worse than test–retest (both movement tasks). Female players showed worse movement quality than males with greater peak values on the frontal and transverse planes at the knee, pelvis and trunk and smaller knee and trunk flexion (d = 0.50–1.1 in Agility T-test, = 0.39–0.73 in FS deceiving action). The on-field cut maneuver kinematics by wearable sensors demonstrated sufficient reliability for most joints. Reliability and normative values might help to objectify ACL injury prevention programs in football academies.

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Rhabdomyolysis due to nutritional supplements intake: Case report. Apunts Sports Medicine, 60: 100479, 2025.

Abstract - Rhabdomyolysis is a condition characterized by the destruction of muscle cells and subsequent release of it's intracellular contents. It is typically asymptomatic, with moderate creatine kinase elevation. Severe cases can involve disseminated intravascular coagulation, liver failure, or acute kidney injury, with a mortality rate of up to 50 %. Nutritional supplements containing compounds such as synephrine, caffeine, and clenbuterol have been reported to precipitate rhabdomyolysis. This case describes a 26-year-old male who, after consuming a full jar of fat burners, developed severe abdominal pain, choluria, myalgias, and muscle weakness. He was diagnosed with rhabdomyolysis and acute renal failure, requiring intensive care unit admission. The unregulated use of nutritional supplements may be an underestimated risk factor for rhabdomyolysis, especially given the fact that many cases are underdiagnosed.

Salary dispersion and team success: The impact of positional spending on win percentage in the National Football League. International Journal of Business in Sports, Tourism & Hospitality Management, 6(1):1-27, 2025.

Abstract - As the NFL's salary cap has surged from $34 million in 1994 to $255 million in 2024, effective cap management has become crucial. However, relatively little focus has been paid to how spending on specific positions affects team success. The purpose of this study was to examine the relationship between salary cap allocation and team success in the NFL, specifically focusing on how spending in different positions affects win percentage. We utilized a quantitative design, analyzing 384 NFL team observations from the 2013–2024 seasons, with linear regression models applied to assess how salary cap allocations across key positions impacted team success, using win percentage as the primary outcome variable and the number of wins as a robustness check. Our findings show that higher spending on quarterbacks, wide receivers, and offensive linemen is strongly linked to improved team performance, while expenditures on running backs and tight ends have little to no significant impact. These results suggest that teams could benefit from reallocating resources to more critical positions,

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particularly quarterbacks and wide receivers, to maximize on-field performance. This study also offers practical insights for NFL teams seeking to refine their salary cap strategies.

Salivary biomarkers of tactical athlete readiness: A systematic review. PLOS ONE, 20(4): e0321223, 2025.

Abstract - Tactical athletes must maintain high levels of physical and cognitive readiness to handle the rigorous demands of their roles. They frequently encounter acute stressors like sleep deprivation, muscle fatigue, dehydration, and harsh environmental conditions, which can impair their readiness and increase the risk of mission failure. Given the challenging conditions these athletes face, there is a vital need for non-invasive, rapidly deployable point-of-care assessments to effectively measure the impact of these stressors on their operational readiness. Salivary biomarkers are promising in this regard, as they reflect physiological changes due to stress. This systematic review aims to investigate salivary markers as potential indicators for readiness, specifically focusing on their sensitivity to acute stressors like sleep deprivation, dehydration, environmental factors, and muscle fatigue. A search was conducted using the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) guidelines (PROSPERO; registration #: CRD42022370388). The primary inclusion criteria were the use of a quantitative analysis to assess salivary biomarkers changes in response to acute stressors. Risk of bias and methodological quality were evaluated with the modified Downs and Black checklist. Hormonal salivary biomarkers were the most commonly studied biomarkers. Muscle damage and fatigue were the most frequently studied acute stressors, followed by sleep deprivation, multiple stressors, dehydration, and environmental. Biomarkers such as creatine kinase, aspartate aminotransferase, uric acid, cortisol, testosterone, and the testosterone to cortisol ratio were indicative of muscle damage. Dehydration influenced osmolality, total protein, flow rate, and chloride ion concentrations. Sleep deprivation affected proteins, peptides, and alpha-amylase levels. Environmental stressors, such as hypoxia and cold temperatures, altered cortisol, pH, dehydroepiandrosterone-sulfate (DHEA-s), and salivary IgA levels. The current body of research highlights that various salivary biomarkers react to acute stressors, and proteomic panels appear promising for predicting physical and cognitive outcomes relevant to the operational readiness of tactical athletes.

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Scoping review of methods of monitoring acute changes in lower body neuromuscular function via force plates. PLOS ONE, 20(5): e0322820, 2025.

Abstract - Force plates are amongst the most utilized technological apparatus for monitoring acute changes in neuromuscular function in sports. Practitioners apply monitoring strategies to manage neuromuscular fatigue and physical preparedness with valid, reliable, and sensitive measures. The aim of this scoping review was to identify, map, and describe the variety of monitoring procedures which have been previously applied in research (e.g., test and metric selection, data collection, study design, and data analysis procedures). Searches were completed by 24th June 2024. One thousand, nine hundred, and seventy-eight studies were identified across four databases (PubMed, EBSCO, Clarivate web of science, and Ovid). After removing duplicates, applying the inclusion criteria, and scouring the reference lists of remaining studies, a final total of thirty studies of within-group repeated measures design were used in this review. Major differences were identified across all aspects of studies methodologies, such as in subject demographics (e.g., sex, sport, and competitive level), data collection protocols (e.g., force plate hardware utilized, test and metric selection, verbal cues, and provision of information regarding testing surface, familiarization and warm-up provided, the process of zeroing force plates between trials, and weighing of subjects during trials), and study design (e.g., reference physical activity investigated, time of season, testing timepoints, and training load determination). A general lack of reporting and uniformity in metric definitions, metric calculations, and phase terminology was identified across studies. For example, two separate calculations were reported for "peak force" across studies, as either "the maximum force achieved throughout the entirety of the trial", or as "the maximum force achieved during the propulsion phase". The latter calculation was also utilized for "peak concentric force" in a separate study. Thus, an accurate comparison of results across studies (e.g., via meta-analysis) and forming any generalized conclusions about the application of specific tests and metrics for monitoring acute changes in neuromuscular function using force plates was premature at this time. The information presented in this review will contribute towards forming a rationale for the data collection, study design, and data analysis protocols for future research on monitoring acute changes in neuromuscular function using force plates.

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Sleep architecture is altered with travel and matches in professional rugby union players. Journal of Sports Sciences, Ahead of Print: 1-9, 2025.

Abstract- Field-based athlete research has primarily focused on sleep-wake patterns, but technological advancements now enable detailed assessment of sleep architecture. This study compared sleep quantity, quality, and architecture during home (no travel; HOME) and away (eastbound travel across three time zones; AWAY) matches in professional male rugby union. The sleep of twenty athletes was assessed using homebased polysomnography over three nights per condition: two nights preceding the match (MD-2), match night (MD), and the following night after (MD + 1). Sleep duration, efficiency, onset/offset, latency, wake after sleep onset, awakenings, and sleep stages (as light, deep and rapid eye movement [REM] sleep, evaluated as proportion [%] and time [min]) were monitored. Compared to HOME, AWAY increased sleep onset latency (+12 ± 23 min; $p < 0.01$) and reduced light sleep (−6.2 ± 7.1%; $p < 0.01$). Match days (MD) incurred delayed bedtime (+71 ± 135 min; $p < 0.001$), reduced sleep duration (−94 ± 120 min; $p < 0.05$), and altered sleep architecture (deep: +9.8 ± 10.0%; REM: −6.9 ± 8.3%; both $p < 0.05$) compared to MD-2. On MD + 1, sleep duration (+96 ± 147 min; $p < 0.01$) and light and REM sleep time rebounded (+52 ± 44 min and + 39 ± 52 min, respectively; both $p < 0.05$). Travel and matches altered sleep architecture in professional rugby. Training and flight schedules should ensure sufficient sleep opportunity after travel and matchday to optimize recovery.

Stingers – A review of current understanding and management. Applied Sciences, 15: 3510, 2025.

Simple Summary: Sports injury can be of multiple sorts. One of the various types is compression or stretch injury to a network of nerves in the neck area. This injury is commonly known as 'Stinger' or 'Burner'. In this review, we summarized the recent understanding and management principles of a stinger. The injury happens to be common in contact sports like American Football and Rugby although Mixed Martial Arts (MMA) can also cause it. It is reported up to 50.3% of football players have been affected. It manifests with numbness, pins and needles sensation, and/or weakness of upper extremities right after a sports incident. There can also be neck stiffness. Mostly it is mild and self-limiting, but in certain instances active management with physiotherapy and rehabilitation is necessary after

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appropriate examinations and diagnostic tests. Cervical x-ray and Magnetic Resonance Imaging (MRI) are the main stays of evaluation even though nerve conduction tests can be utilized. In rare and unlikely instances where the nerve(s) gets severed, surgical intervention is needed and recovery may not be full. However, most athletes return to their regular engagement in sports, often not missing the next practice or game.

Abstract: A stinger is the transient sensory and motor loss of one upper extremity caused by a stretch or compression injury to the brachial plexus or the exiting cervical nerve roots. Athletes from contact sports like American football, rugby, wrestling, and horseback riding are most affected. Given today's competitive athletic culture and recent advancements in sports medicine, understanding the clinical, diagnostic, and therapeutic aspects of stingers is crucial. Thus, this narrative review highlights current knowledge of pathology, clinical features, diagnostic modalities, radiologic measurements, management, rehabilitation, and return-to-play protocols. Recent and prominent English publications on adult athletes revealed that the brachial plexus upper roots (C5 and C6) are most affected. Stingers accounted for 62.2% of neck injuries in National Football League athletes and affected up to 50.3% of football players. Grading is based on injury severity to the nerves. Most stingers are mild, lasting a few seconds to minutes. Return-to-play protocols remain controversial; however, stingers have promising prognoses and can mostly be resolved with conservative treatment and rehabilitation. Severe injuries require surgical intervention but rarely occur in athletics. In conclusion, many recent developments in the evaluation and treatment of stingers impact current treatment plans, return-to-play decisions, and the future of athletes.

Systematic review of motion capture in virtual reality: Enhancing the precision of sports training. Journal of Ambient Intelligence and Smart Environments, 17(1): 5-27, 2025.

Abstract - In the modern era of sports training, the synergy between motion capture and Virtual Reality (VR) offers an innovative approach to enhancing training precision. This systematic review delves into the application of motion capture within VR for sports training, highlighting its transformative potential. Through a comprehensive literature search, we examined the myriad applications, from physical

conditioning enhancements to accelerated rehabilitation processes. Our findings underscore the capability of real-time feedback, immersive training environments, and tailored regimes that this fusion provides. However, despite its promise, challenges such as hardware constraints, data processing complexities, and interaction interface limitations persist. Future trajectories indicate an increasing influence of AI and deep learning, promising more sophisticated hardware and a broader spectrum of applications, including niche sports disciplines. The review concludes with an emphasis on the wider societal implications, suggesting a shift towards a holistic athlete wellbeing approach.

The Deceleration Dilemma: Leveraging the relationship between quadriceps function and deceleration to optimize return to sport. International Journal of Sports Physical Therapy, 20(5): 762-765, 2025.

Abstract - Decelerations frequently occur during sport and are often associated with non-contact knee injury mechanisms. Though decelerations place a high eccentric demand on the quadriceps, little is known about the relationship between quadriceps function and deceleration performance. Traditional functional testing batteries often over-estimate quadriceps function and lack assessments of deceleration performance, while current deceleration testing is often heavily reliant on technology and performed at shallow angles. The purpose of this clinical commentary is to describe the relationship between deceleration and quadriceps function, suggest two clinical tests to assess deceleration performance, and provide exercise suggestions for improving deceleration performance. The authors believe that assessing a deceleration task and a single leg retro hop may be a viable alternative for assessing the quadriceps when isokinetic or isometric testing may not be available. Level of Evidence: 5

The Doctor is (Not) In: Implications of Increased Medical Malpractice Litigation on Team Physicians in the National Football League, 32 Jeffrey S. Moorad Sports Law Journal, 109: 1-46, 2025. *No abstract available.*

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The effect of fatiguing muscle contractions on kicking performance in experienced soccer players. Sports Biomechanics, Ahead of Print: 1-21, 2025.

Abstract - The objective of this study was to clarify the effects of fatiguing muscle contractions of the m. quadriceps femoris on kicking abilities of experienced soccer players. 16 male professional (n = 5) and amateur players (n = 11) performed kicking tests in two conditions (fatigue and control) on separate days in a randomized crossover design. The fatiguing protocol performed with the kicking leg consisted of 5 sets of 10 maximal voluntary concentric and eccentric knee extensions. Maximal voluntary isometric contraction force (MVIC), 15 hz/50 hz stimulation force ratio (force ratio), and kicking abilities were assessed before and after completion of the fatiguing protocol or rest (control). The fatiguing protocol successfully induced fatigue of 14.0 ± 2.7% (mean ± SE) reduced MVIC and 14.0 ± 3.7% reduced force ratio while no reductions occurred in the control condition. Between group difference showed ball speed declined 2.1 ± 0.95% more following the fatigue protocol compared to control condition. On the control day shooting accuracy improved by 13.3 ± 5.6% and was numerically impaired on the intervention day by 1.0 ± 9.2%. Despite this, no significant between group difference was observed in shooting accuracy (p = 0.18). The study demonstrated that fatigue induced by prior muscle contractions impairs maximal shooting speed, but we observed no significant impairment of shooting accuracy.

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